

NEURAL NETWORKS: How Computers Learn Like Our Brain

Teacher Lesson Plan | Intro to AI & Neural Networks

Grade Level 3rd – 6th grade	Duration 50–60 minutes	Subject Science / Technology / Intro to AI	Group Size Whole class, with pair/share moments
---------------------------------------	----------------------------------	---	--

Learning Objectives

By the end of this lesson, students will be able to:

- Explain, in simple terms, what a neuron is and how neurons send messages in the brain.
- Describe the three main parts of a neuron: dendrites, axon, and synapse.
- Explain the analogy between the human brain and an artificial neural network.
- Identify at least three everyday technologies that use neural networks (e.g., face recognition, voice assistants, recommendation systems).
- Describe, in basic terms, how a neural network "learns" from examples.

Materials Needed

- Projector or screen to display slides/images (brain, neuron diagram, phone face-unlock, smart speaker, etc.)
- Whiteboard or chart paper and markers
- Sticky notes or index cards (one per student) for the brainstorm activity
- Optional: tablets/computers with internet access for Google Teachable Machine or Machine Learning for Kids demo

Key Vocabulary

Term	Student-Friendly Definition
Neuron	A tiny messenger cell in the brain that helps you think, learn, and remember.
Dendrite	The part of a neuron that catches incoming messages.
Axon	The part of a neuron that sends messages to other neurons.
Synapse	A tiny bridge where one neuron passes a message to another.
Neural Network	A "digital brain" inside a computer, made of connected parts that work together like neurons.
Training	Showing a computer lots of examples so it can learn to recognize patterns.
AI (Artificial Intelligence)	Technology that lets computers learn and make decisions, a bit like a brain does.

Lesson Sequence

Time	Segment	What Happens
5 min	Hook: "Can a Computer Think?"	Ask: "Do you think a computer could ever learn the way you do?" Take a few quick answers, no wrong responses. Introduce today's big question: how computers learn using something called a <i>neural network</i> , modeled after our own brains.
8 min	What Is Your Brain? / Meet the Neuron	Explain that the brain is made of billions of neurons that send messages to each other. Show the neuron diagram and walk through its three parts: dendrites (catch messages), axon (sends messages), synapse (the bridge to the next neuron). Have students wiggle their fingers like "dendrites catching a message."
5 min	How Your Brain Learns	Discuss how practicing something (like riding a bike) strengthens the connections between neurons. Ask students to share a skill they got better at with practice and connect it to neurons "talking" more.
7 min	What Is a Neural Network?	Introduce the brain-vs-computer comparison: a neural network is a "digital brain" with connected parts, just like neurons. Emphasize that computers learn from examples, the same way students learn from practice — the more examples, the smarter it gets.

Time	Segment	What Happens
15 min	Real-World Examples	Walk through four everyday examples, pausing for quick discussion after each: • Face recognition — how a phone camera checks facial features to unlock. • Voice assistants (Alexa, Siri, Google Assistant) — how they turn sound waves into words. • Recommendation systems (YouTube/Netflix) — how watching videos teaches the system your interests. • Smart cameras — how they label animals, vehicles, and objects in a photo. Ask students which of these they have personally used.
5 min	Training a Neural Network	Explain the "1,000 cat pictures" example: a computer learns what a cat looks like by studying many examples, just like a student learns by practicing again and again.
8 min	Fun Activity: AI Brainstorm	Hand out sticky notes/index cards. Students write or draw one idea for a problem AI could help solve in the future. Remind them there are no wrong answers. Volunteers share their ideas with the class; post notes on a "Future AI Wall."
2 min	Wrap-Up & Resources	Recap the brain-to-network analogy. Share where curious students can explore further: Scratch AI blocks, the Machine Learning for Kids website, and Google Teachable Machine. Close with: "Your curiosity is your superpower!"

Discussion Questions

- What is one job that neurons do inside your brain?
- How is a neural network similar to your brain? How is it different?
- Why does a computer need lots of examples (like 1,000 cat pictures) to learn something?
- Can you name a device or app you use that has a neural network inside it?
- What is one new thing you wish a computer could learn to help you with?

Assessment Ideas

- Exit ticket: have students draw and label a simple neuron (dendrites, axon, synapse) and write one sentence about how it relates to neural networks.
- Listen for accurate use of vocabulary (neuron, neural network, training) during class discussion.
- Review the "Future AI Wall" sticky notes for evidence that students understood the learn-from-examples concept.

Extension for older or advanced students: Try a guided, teacher-led demo of Google Teachable Machine or the Machine Learning for Kids website, training a simple model live in class on a webcam or sound example to make the "training" concept concrete.